

Counting $Z\varphi$ -classes for PerfectGroup(7680, 1)

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n	$ S $	$ E $	#SG($8n$)	# $Z\varphi$	cumulative	time (s)	cum. time (s)
2	16	15 360	14	11	11	0.959	0.959
3	24	23 040	15	1	12	0.515	1.474
4	32	30 720	51	103	115	6.804	8.278
5	40	38 400	14	1	116	0.594	8.872
6	48	46 080	52	22	138	5.091	13.963
7	56	53 760	13	1	139	0.467	14.430
8	64	61 440	267	1 065	1 204	68.931	83.361
9	72	69 120	50	2	1 206	2.793	86.154
10	80	76 800	52	22	1 228	4.669	90.823
11	88	84 480	12	1	1 229	0.442	91.265
12	96	92 160	231	334	1 563	53.079	144.344
13	104	99 840	14	1	1 564	0.535	144.879
14	112	107 520	43	22	1 586	5.157	150.036
15	120	115 200	47	1	1 587	2.280	152.316
16	128	122 880	2328	16 823	18 410	1649.469	1801.785
17	136	130 560	15	1	18 411	1.055	1802.840
18	144	138 240	197	55	18 466	25.442	1828.282
19	152	145 920	12	1	18 467	0.495	1828.777
20	160	153 600	238	347	18 814	53.798	1882.575
21	168	161 280	57	2	18 816	3.263	1885.838
22	176	168 960	42	22	18 838	5.161	1890.999
23	184	176 640	12	1	18 839	0.569	1891.568
24	192	184 320	1543	5 250	24 089	760.702	2652.270
25	200	192 000	52	2	24 091	2.932	2655.202
26	208	199 680	51	22	24 113	4.950	2660.152
27	216	207 360	177	5	24 118	14.355	2674.507
28	224	215 040	197	330	24 448	49.071	2723.578
29	232	222 720	14	1	24 449	0.517	2724.095
30	240	230 400	208	44	24 493	22.509	2746.604
31	248	238 080	12	1	24 494	0.511	2747.115
Total				24 494	24 494	2747.115	2747.115

Values of n contributing the largest shares of the total runtime

n	time (s)	share of total time	# $Z\varphi$	share of all $Z\varphi$ -classes
16	1649.469	60.04%	16 823	68.68%
24	760.702	27.69%	5 250	21.43%
8	68.931	2.51%	1 065	4.35%
20	53.798	1.96%	347	1.42%
Top four combined		92.20%		95.88%

Audited memory-optimized count-only computation, $2 \leq n \leq 31$

Structural and computational remarks. $\text{Out}(P) \simeq V_4$ acts faithfully on $Z(P) \simeq C_4 \times C_2$; equivalently, the kernel of the action is trivial. The auxiliary group S has order $8n$, and the resulting extensions have order $7680n$. All values in the first table were obtained with the same audited **count-only** implementation. The implementation is optimized primarily for memory: after a SmallGroup and the relevant subgroups/orbits have been processed, the constructed group objects are discarded rather than retained in a global list; only compact counting data are accumulated.

The second table shows that the runtime is strongly concentrated on the values of n that also contribute most of the $Z\varphi$ -classes. For the four largest runtime contributors together, the share is 92.20% of the total measured runtime and 95.88% of all $Z\varphi$ -classes for $2 \leq n \leq 31$.